6-1 Journal: Emerging Technology and Artifact Update

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Two technologies that are changing the field of computer science dramatically are Cloud Development/Computing and the much newer Quantum Computation. While Cloud Development has been used for some time, Quantum Computers are very new and unproven technology that hold a lot of promises. Both technologies will continue to alter how we design our future system's functionality, scalability, and security.

Cloud computing as a concept has been around since the 60's, but real cloud computing and Software-as-a-service programs were first used in markets in 1999 with Salesforce then later in 2006 with AWS. Cloud Computing is essentially a model that gives developers on demand access to rent and utilize other companies' servers, databases, and storage over an internet connection. This allows developers not to worry about buying, running, maintaining, or securing their own server farms which save developers lots of time and money. AWS and Salesforce are both used heavily in the web development space to this day, along with other competitors such as Microsoft Azure and Google Cloud. Most programs that you use everyday like Spotify, Netflix, eBay, and Uber are all hosted on cloud services. This has changed the computer science field by forcing developers to get more familiar with developing in cloud development suites and forces

applications to be designed for more distributed environments and be horizontally scalable. Cloud computing will continue to evolve as hardware components become more difficult for average consumers to buy. This means newer services like Cloud Gaming or cloud AI suites will become unfortunately more necessary for consumers to use.

Quantum Computing, while technically achieved in 1998, is still very unproven in comparison to Cloud Computing. In short, Quantum Computing works completely differently from how modern computers do as they use qubits to process information as opposed to binary bits. So, while modern computers only compute through 1s and 0s, Quantum computers can have qubits that exist in the 0 state, 1 state, or both simultaneously. This allows the system to solve problems millions of times faster than a classic supercomputer can (Tepanyan, January 29, 2025). While this technology would revolutionize our computing abilities it would also threaten the security of our current systems and infrastructure. This is because modern cryptography algorithms that we use could become solvable by quantum computers, leaving entire systems exposed. While researchers have begun to explore options for security in a post quantum-computing world, there is still much to be discussed and solved for future systems to be secure and reliable. On the bright side, once quantum computers do become more utilizable, our advancements in every field would accelerate significantly such as our developments in medicine, climate change, astronomy, theoretical physics, etc. Quantum Computing will be game-changing for our world and will help our society solve many of the issues facing us today, but we will need to ensure that the security risks that come along with them are mitigated as best as possible.

For my project, I have it in a near complete state. I have successfully converted the program from Java to C++ to give the program better performance and allow for more direct

control over the memory allocation of the program. I also changed the way my program assigns and sorts contact IDs on the memory layer of my program. I did this by changing my program from using an Array list with a 0,1,2,… form to a faster and more secure hash system. This allows for contacts to be found in memory faster while also preventing an issue from occurring where contacts may be assigned to the same ID. Lastly, my system did not have a database to store memory, so I implemented a brand-new database that uses SQLite to store contacts persistently, making them accessible after runtime. There are still some last few changes that need to be made specifically to my program's hash function that need to be made to prevent a few bugs from occurring, but I am very happy with where I have succeeded so far throughout this course.

Citations:

Tepanyan Hayk. "Quantum Computing vs. Classical Computing." Quantum Computing Vs. Classical Computing. January 29, 2025. https://www.bluequbit.io/blog/quantum-computing-vs-classical-computing.